

starting functions $f_j(x)$ and $f_{j+1}(x)$: Stability is a property of the recurrence, not of the starting values.

An alternative heuristic procedure for testing stability is to replace the recurrence relation by a similar one that is linear with constant coefficients. For example, the relation (5.4.2) becomes

$$y_{n+1} - 2\gamma y_n + y_{n-1} = 0 \quad (5.4.11)$$

where $\gamma \equiv n/x$ is treated as a constant. You solve such recurrence relations by trying solutions of the form $y_n = a^n$. Substituting into the above recurrence gives

$$a^2 - 2\gamma a + 1 = 0 \quad \text{or} \quad a = \gamma \pm \sqrt{\gamma^2 - 1} \quad (5.4.12)$$

The recurrence is stable if $|a| \leq 1$ for all solutions a . This holds (as you can verify) if $|\gamma| \leq 1$ or $n \leq x$. The recurrence (5.4.2) thus cannot be used, starting with $J_0(x)$ and $J_1(x)$, to compute $J_n(x)$ for large n .

Possibly you would at this point like the security of some real theorems on this subject (although we ourselves always follow one of the heuristic procedures). Here are two theorems, due to Perron [2]:

Theorem A. If in (5.4.9) $a_n \sim an^\alpha$, $b_n \sim bn^\beta$ as $n \rightarrow \infty$, and $\beta < 2\alpha$, then

$$g_{n+1}/g_n \sim -an^\alpha, \quad f_{n+1}/f_n \sim -(b/a)n^{\beta-\alpha} \quad (5.4.13)$$

and f_n is the minimal solution to (5.4.9).

Theorem B. Under the same conditions as Theorem A, but with $\beta = 2\alpha$, consider the *characteristic polynomial*

$$t^2 + at + b = 0 \quad (5.4.14)$$

If the roots t_1 and t_2 of (5.4.14) have distinct moduli, $|t_1| > |t_2|$ say, then

$$g_{n+1}/g_n \sim t_1 n^\alpha, \quad f_{n+1}/f_n \sim t_2 n^\alpha \quad (5.4.15)$$

and f_n is again the minimal solution to (5.4.9). Cases other than those in these two theorems are inconclusive for the existence of minimal solutions. (For more on the stability of recurrences, see [3].)

How do you proceed if the solution that you desire *is* the minimal solution? The answer lies in that old aphorism, that every cloud has a silver lining: If a recurrence relation is catastrophically unstable in one direction, then that (undesired) solution will decrease very rapidly in the reverse direction. This means that you can start with *any* seed values for the consecutive f_j and f_{j+1} and (when you have gone enough steps in the stable direction) you will converge to the sequence of functions that you want, times an unknown normalization factor. If there is some other way to normalize the sequence (e.g., by a formula for the sum of the f_n 's), then this can be a practical means of function evaluation. The method is called *Miller's algorithm*. An example often given [1,4] uses equation (5.4.2) in just this way, along with the normalization formula

$$1 = J_0(x) + 2J_2(x) + 2J_4(x) + 2J_6(x) + \cdots \quad (5.4.16)$$

Incidentally, there is an important relation between three-term recurrence relations and *continued fractions*. Rewrite the recurrence relation (5.4.9) as

$$\frac{y_n}{y_{n-1}} = -\frac{b_n}{a_n + y_{n+1}/y_n} \quad (5.4.17)$$

Iterating this equation, starting with n , gives

$$\frac{y_n}{y_{n-1}} = -\frac{b_n}{a_n - \frac{b_{n+1}}{a_{n+1} - \dots}} \quad (5.4.18)$$

Pincherle's theorem [2] tells us that (5.4.18) converges if and only if (5.4.9) has a minimal solution f_n , in which case it converges to f_n/f_{n-1} . This result, usually for the case $n = 1$ and combined with some way to determine f_0 , underlies many of the practical methods for computing special functions that we give in the next chapter.

5.4.2 Clenshaw's Recurrence Formula

Clenshaw's recurrence formula [5] is an elegant and efficient way to evaluate a sum of coefficients times functions that obey a recurrence formula, e.g.,

$$f(\theta) = \sum_{k=0}^N c_k \cos k\theta \quad \text{or} \quad f(x) = \sum_{k=0}^N c_k P_k(x)$$

Here is how it works: Suppose that the desired sum is

$$f(x) = \sum_{k=0}^N c_k F_k(x) \quad (5.4.19)$$

and that F_k obeys the recurrence relation

$$F_{n+1}(x) = \alpha(n, x) F_n(x) + \beta(n, x) F_{n-1}(x) \quad (5.4.20)$$

for some functions $\alpha(n, x)$ and $\beta(n, x)$. Now define the quantities y_k ($k = N, N - 1, \dots, 1$) by the recurrence

$$\begin{aligned} y_{N+2} &= y_{N+1} = 0 \\ y_k &= \alpha(k, x) y_{k+1} + \beta(k, x) y_{k+2} + c_k \quad (k = N, N - 1, \dots, 1) \end{aligned} \quad (5.4.21)$$

If you solve equation (5.4.21) for c_k on the left, and then write out explicitly the sum (5.4.19), it will look (in part) like this:

$$\begin{aligned} f(x) &= \dots \\ &+ [y_8 - \alpha(8, x) y_9 - \beta(9, x) y_{10}] F_8(x) \\ &+ [y_7 - \alpha(7, x) y_8 - \beta(8, x) y_9] F_7(x) \\ &+ [y_6 - \alpha(6, x) y_7 - \beta(7, x) y_8] F_6(x) \\ &+ [y_5 - \alpha(5, x) y_6 - \beta(6, x) y_7] F_5(x) \\ &+ \dots \\ &+ [y_2 - \alpha(2, x) y_3 - \beta(3, x) y_4] F_2(x) \\ &+ [y_1 - \alpha(1, x) y_2 - \beta(2, x) y_3] F_1(x) \\ &+ [c_0 + \beta(1, x) y_2 - \beta(1, x) y_2] F_0(x) \end{aligned} \quad (5.4.22)$$